

FINANCING RESILIENCE AND WEFE NEXUS PROFESSIONAL SERIES

EGYPT COUNTRY REPORT

# Integrated Desalination and Reuse of Agricultural Drainage Water

## for Food Security, Renewable Energy, and Ecosystem Resilience

*Grounded in Egypt's operating mega-plants — Bahr El-Baqar, Al-Mahsama and New Delta (El-Hammam)*

Prepared by

**Dr. Hassan Tolba Aboelnga**

*Water Security & WEFE Nexus Specialist | TH Köln, Institute for Technology and Resources Management in the Tropics and Subtropics (ITT) | Chair, IWRA Water Security Task Force*

Prepared for institutional, donor and investor audiences

Sector: Water–Energy–Food–Ecosystems (WEFE) Nexus | Geography: Arab Republic of Egypt

Training Series Reference Document | July 2026

## Preface — Purpose and Use of This Document

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This report is issued as the reference case study of the Financing Resilience and WEFE Nexus Professional Series. It is written to serve two audiences simultaneously: first, as a standalone investment-grade project report on drainage-water reuse in Egypt; and second, as a worked template that trainees and project teams can follow, section by section, when preparing WEFE Nexus project reports for other countries, sectors, or sites.

Every section below follows a fixed structure — narrative framing, quantified evidence, a standard table or matrix, and a scored conclusion — so that the same 24-section skeleton can be reused for a different project simply by replacing the underlying data. Where this report draws on a real, operating Egyptian programme (the Bahr El-Baqar, Al-Mahsama and New Delta / El-Hammam plants) rather than a hypothetical, that choice is deliberate: it demonstrates how to ground each section of the template in verifiable, cited evidence instead of generic placeholder text, which is the standard trainees should apply to their own case studies.

A short methodological note on replication is provided in the final section of this report, 'Using This Template for Other Projects,' which distills the drafting logic behind each of the 24 sections into a reusable checklist.

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# 1. Executive Summary

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Egypt's freshwater endowment has fallen to roughly 560 m<sup>3</sup> per capita per year — well below the 1,000 m<sup>3</sup> water-poverty threshold and closer still to the 500 m<sup>3</sup> absolute-scarcity line — as the Nile's fixed annual allocation of about 55.5 billion cubic metres (BCM) is stretched across a population that has passed 106 million. Agricultural drainage water, historically treated as a disposal problem, has become one of the country's few expandable water resources: the drainage network carries in excess of 14 BCM per year, most of it discharged untreated into lakes, drains and the Mediterranean.

Rather than a hypothetical pipeline concept, this report is built around a programme Egypt is already constructing at national scale. Between roughly 2018 and 2025 the Ministry of Water Resources and Irrigation (MWRI), together with the Armed Forces Engineering Authority, commissioned three mega-plants that together define the country's drainage-water-reuse model:

- Bahr El-Baqar WWTP (Sinai / east of the Suez Canal) — treatment capacity of about 5.6 million m<sup>3</sup>/day, supporting reclamation of roughly 456,000 feddans.
- Al-Mahsama WWTP (Ismailia, east bank of the Suez Canal) — capacity of about 0.365 BCM/year, supporting reclamation of roughly 70,000 feddans.
- New Delta / El-Hammam WWTP (Western Desert) — the largest of the three and among the largest water-treatment plants in the world, with a design capacity in the range of 2.2 BCM/year (roughly 6 million m<sup>3</sup>/day), supporting reclamation targeted at up to 1.5 million feddans.

Egyptian officials have indicated that, taken together, these three plants are expected to deliver on the order of 5 BCM of treated drainage water per year by 2026 — a volume comparable to a major new river allocation, generated entirely from water Egypt was previously discharging as waste. This report uses that operating programme as the evidence base for a fourth, replicable project archetype: decentralised, solar-powered drainage-water desalination and reuse units that can be deployed across the Nile Delta's secondary drainage network to extend the same logic to medium-scale, farmer-facing infrastructure.

The proposed project — sized here at USD 670 million in an illustrative configuration — combines drainage water collection, reverse-osmosis desalination, 150 MW of solar generation, an irrigation distribution network, constructed wetlands and a digital water-management platform. It is structured for blended finance, aligns directly with Egypt Vision 2030 and the National Climate Change Strategy 2050, and is assessed here as investment-ready pending a full feasibility study, with a WEFE Nexus composite score of 95/100.

Three findings distinguish this report from a conventional pre-feasibility note. First, the technology and delivery risk are already substantially retired: Egypt has proven, at national scale and within a single decade, that mega-scale drainage treatment and inter-basin conveyance (including the New Delta project's underground pipeline system carrying water tens of kilometres into the Western Desert) can be engineered, financed and operated. Second, the resource itself is abundant and currently undervalued: at over 14 BCM/year of total drainage flow, the volume already treated by the three flagship plants — on the order of 5 BCM/year by 2026 — still represents well under half of the theoretically recoverable stream, leaving substantial headroom for a decentralised second wave of investment such as the project proposed here. Third, the social and environmental risk profile is well documented rather than speculative: independent field research on Al-Mahsama has already identified the specific mechanism by which drainage reallocation can affect old-land smallholders, which allows this report to design safeguards against a known risk rather than a hypothetical one.

The remainder of this report is organised as a full, investable project document — background, WEFE Nexus analysis, technical description, stakeholder mapping, climate resilience, logframe, screening and bankability, financials, capital structure, risk, funding-call fit, MRV, safeguards, expected impacts, readiness, investor pitch

and conclusions — followed by a closing methodological section explaining how the same structure should be applied to other WEFE Nexus projects in the training series.

## 2. Strategic Assessment

### 2.1 Project Classification

| Attribute           | Detail                                                                                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Type        | Climate-Resilient WEFE Nexus Infrastructure — decentralised drainage water desalination and reuse                                                             |
| Country             | Arab Republic of Egypt                                                                                                                                        |
| Reference Geography | Nile Delta secondary drainage network, benchmarked against Bahr El-Baqar (Sinai), Al-Mahsama (Ismailia/Suez Canal) and New Delta / El-Hammam (Western Desert) |
| Programme Status    | Reference plants operational or in advanced commissioning (2018–2025); this project extends the model to new sites                                            |

### 2.2 Composite Assessment Scores

| Criterion                | Score (/10) |
|--------------------------|-------------|
| Water Impact             | 10          |
| Energy Integration       | 9           |
| Food Security Impact     | 10          |
| Ecosystem Benefits       | 8           |
| Climate Resilience       | 10          |
| Financial Attractiveness | 8           |
| Innovation               | 9           |
| Replicability            | 10          |

### 2.3 Recommendation

High-priority investment candidate. Egypt's demonstrated ability to deliver Bahr El-Baqar, Al-Mahsama and New Delta at gigawatt-and-billion-cubic-metre scale substantially de-risks execution for follow-on, decentralised projects of this kind. Strong candidate for:

- Green Climate Fund (GCF)
- Adaptation Fund
- European Investment Bank (EIB)
- European Bank for Reconstruction and Development (EBRD)
- Islamic Development Bank (IsDB)
- World Bank

- EU Global Gateway

### 3. Project Background and Rationale

Egypt receives under 600 m<sup>3</sup> per capita per year of water, against a Nile allocation fixed by the 1959 agreement at 55.5 BCM/year for Egypt — a volume unchanged for over six decades while population has more than doubled. The Grand Ethiopian Renaissance Dam (GERD) filling has added further hydrological uncertainty upstream. Key structural challenges include water scarcity, seawater and drain-water salinity intrusion into the Delta aquifer, climate-driven variability, stagnating agricultural productivity per unit of water, and rising energy costs for conventional treatment.

Egypt's response, branded domestically as Irrigation System 2.0, has centred on treating agricultural drainage water as a resource rather than a waste stream. The three flagship plants already reallocate treated drainage water from the Delta's 'old lands' to newly reclaimed desert-fringe 'new lands' in Sinai, around the Suez Canal, and in the Western Desert — a model this report proposes to replicate at decentralised, solar-integrated scale closer to the point of drainage generation.

The engineering logic of the New Delta / El-Hammam scheme illustrates the scale of ambition involved: agricultural wastewater that would historically have discharged through the Rosetta branch of the Nile into the Mediterranean is instead redirected roughly 50 kilometres along the coast, carried underground through large-diameter pipelines for a further 22 kilometres into the Western Desert, and then conveyed by canal another 42 kilometres to the treatment plant itself, from where treated water irrigates reclaimed land at a reported rate on the order of 6 million m<sup>3</sup>/day at full build-out. Between 2015 and 2020 alone, Egypt reported irrigating more than 5,000 km<sup>2</sup> of previously desert land using this reallocated-drainage model — a scale of land reclamation that positions drainage reuse as a central pillar of national food-security policy rather than a peripheral pilot activity.

This scale of investment is not incidental. It reflects a structural reality: Egypt's Nile allocation has been fixed since the 1959 Nile Waters Agreement while population, urbanisation and irrigated area have all expanded several-fold, and hydrological uncertainty from upstream GERD operations adds a further layer of supply risk that drainage reuse is explicitly designed to buffer against. Understanding this policy driver is essential background for any project team preparing a comparable report for another country: the opening section of the template should always identify the specific structural constraint — a fixed allocation, a transboundary dependency, an aquifer in decline — that makes the proposed WEFE Nexus intervention a policy priority rather than merely a technically interesting option.

#### 3.1 The Existing Egyptian Reference Programme

| Plant                      | Location                                              | Capacity                         | Reclamation Target       |
|----------------------------|-------------------------------------------------------|----------------------------------|--------------------------|
| Bahr El-Baqar WWTP         | East of Suez Canal, Sinai (~10 km south of Port Said) | ≈5.6 million m <sup>3</sup> /day | ≈456,000 feddans         |
| Al-Mahsama WWTP            | East bank of Suez Canal, Ismailia                     | ≈0.365 BCM/year                  | ≈70,000 feddans          |
| New Delta / El-Hammam WWTP | Western Desert, north-west coast                      | ≈2.2 BCM/year (world's largest)  | up to ≈1,500,000 feddans |

Figures compiled from Egyptian government, Armed Forces Engineering Authority and independent academic sources (2023–2025); treat as order-of-magnitude reference values for planning purposes, to be confirmed against the feasibility study for any specific new site.

## 4. Problem Statement

| Challenge                  | Impact                                                                                                |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| Freshwater scarcity        | Reduced irrigation supply per feddan; competition between Delta 'old lands' and reclaimed 'new lands' |
| Salinity increase          | Lower crop productivity, especially in the northern Delta governorates                                |
| Energy-intensive treatment | High operating expenditure for conventional desalination                                              |
| Ecosystem degradation      | Biodiversity loss in Lake Manzala and other drainage-fed wetlands                                     |
| Climate variability        | Reduced resilience of both Nile-fed and drainage-fed irrigation systems                               |

## 5. Project Objectives

### 5.1 Overall Goal

Improve climate-resilient water security and food production in Egypt through the integrated, decentralised reuse of agricultural drainage water, complementing the national Bahr El-Baqar / Al-Mahsama / New Delta programme.

### 5.2 Specific Objectives

1. Recover 200 million m<sup>3</sup>/year of drainage water at the pilot cluster of sites.
2. Produce renewable energy through a co-located 150 MW solar PV array.
3. Irrigate 50,000 hectares of Delta-fringe and reclaimed land.
4. Increase agricultural productivity per unit of water by 20%.
5. Reduce groundwater abstraction in stressed aquifer zones.
6. Improve ecosystem health in receiving drains and wetlands.

## 6. Alignment with Egypt Vision 2030 and National Strategies

### 6.1 Egypt Vision 2030

- Water Security
- Climate Adaptation
- Food Security
- Renewable Energy
- Sustainable Development

### 6.2 National Climate Change Strategy 2050

- Adaptation
- Water Resilience
- Green Growth

### **6.3 Irrigation System 2.0**

The Ministry of Water Resources and Irrigation's Irrigation System 2.0 programme — reviewed by President Abdel Fattah El-Sisi in August 2025 — explicitly foregrounds the Bahr El-Baqar, Al-Mahsama and New Delta plants as its water-treatment backbone, alongside satellite monitoring, digital twins and smart water management. This project is designed as a decentralised, next-tier extension of that same national strategy.

## 7. WEFE Nexus Analysis

| Nexus Sector | Contribution                                                                                            |
|--------------|---------------------------------------------------------------------------------------------------------|
| Water        | Drainage water desalination and reuse, modelled on the Bahr El-Baqar / Al-Mahsama / New Delta programme |
| Energy       | Co-located solar PV generation offsetting desalination OPEX                                             |
| Food         | Irrigation water for reclaimed and Delta-fringe agricultural land                                       |
| Ecosystems   | Wetland restoration and salinity management around receiving drains and lakes                           |
| Enablers     | AI forecasting, digital twin, satellite monitoring — consistent with Irrigation System 2.0              |

### Degree of Integration

#### FULL WEFE INTEGRATION

## 8. Project Description

### Component 1 — Drainage Water Collection System

Networked intake structures on secondary drains, modelled on the Al-Mahsama I/II pumping-station configuration that lifts drainage water from the Delta 'old lands' to treatment sites on higher desert-fringe ground.

### Component 2 — Desalination Facility

**Capacity:** 200 million m<sup>3</sup>/year

**Technology:** Reverse Osmosis (RO), with pre-treatment stages (screening, sedimentation, filtration) as used at Bahr El-Baqar

### Component 3 — Solar Energy System

**Capacity:** 150 MW

### Component 4 — Distribution Network

Irrigation conveyance infrastructure connecting treated water to reclaimed land, comparable in principle to the Sheikh Jaber Canal that receives Bahr El-Baqar's treated effluent.

### Component 5 — Nature-Based Solutions

- Constructed wetlands
- Biodiversity restoration around drainage-fed lakes (e.g., Lake Manzala catchment analogues)
- Carbon sequestration zones

### Component 6 — Digital Water Management Platform

- AI-based demand and flow forecasting
- SCADA control systems
- Digital twin of the drainage-to-irrigation network
- Satellite and drone-based remote sensing, consistent with MWRI's Irrigation System 2.0

## 9. Stakeholder Analysis

| Stakeholder                                                                               | Role                                                                    |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Ministry of Water Resources and Irrigation (MWRI)                                         | Lead agency; owner of the national drainage-reuse strategy              |
| Ministry of Agriculture and Land Reclamation                                              | Beneficiary; land reclamation coordination                              |
| New and Renewable Energy Authority (NREA)                                                 | Renewable energy integration                                            |
| Armed Forces Engineering Authority                                                        | Delivery partner on comparable mega-projects (Bahr El-Baqar, New Delta) |
| Private operators / EPC contractors                                                       | PPP implementation and O&M                                              |
| Farmers' associations (Delta and reclaimed land)                                          | End users of irrigation water                                           |
| Multilateral development banks                                                            | Financing                                                               |
| Universities and research institutes (e.g., TH Köln–Ain Shams–Mansoura iWater consortium) | Research, monitoring and capacity building                              |

## 10. Climate Resilience Assessment

### Climate Risks

- Drought and reduced Nile flow variability, compounded by upstream GERD operations
- Heat waves affecting crop water demand
- Salinity intrusion in the northern Delta and coastal aquifers
- Flooding events affecting drainage infrastructure

### Adaptation Benefits

- Alternative water source independent of the fixed Nile allocation
- Reduced dependency on Nile water for reclaimed land
- Increased agricultural resilience through diversified supply
- Improved food security in a country importing a large share of its staple grain needs

### Resilience Rating

**HIGH**

## 11. Logical Framework (Logframe)

| Intervention Logic                               | Indicators                                     | Means of Verification    | Assumptions / Risks                |
|--------------------------------------------------|------------------------------------------------|--------------------------|------------------------------------|
| Goal: Improve water and food security resilience | Water Security Index +20%                      | National reports         | Stable governance                  |
| Outcome 1: Additional irrigation supply          | 200 million m <sup>3</sup> /year supplied      | Utility data             | Farmer adoption                    |
| Outcome 2: Renewable energy generation           | 150 MW installed                               | Energy reports           | Solar resource performance         |
| Outcome 3: Ecosystem restoration                 | 5,000 ha restored                              | Environmental monitoring | Community support                  |
| Output 1: Desalination plant                     | Plant operational                              | Completion certificate   | Construction completed on schedule |
| Output 2: Solar PV system                        | Energy delivered to grid/plant                 | SCADA records            | Grid/plant integration achieved    |
| Output 3: Digital monitoring system              | Real-time monitoring active                    | Dashboard reports        | Data availability and connectivity |
| Activities                                       | Procurement, construction, operation, training | Progress reports         | Funding secured                    |

## 12. Initial WEFE Screening and Prioritization

| Project                                                         | Sectors Covered | Co-Benefits | Finance Potential | Resilience Impact | Priority |
|-----------------------------------------------------------------|-----------------|-------------|-------------------|-------------------|----------|
| Drainage Water Desalination + Solar (this project)              | W,E,F,Eco       | Very High   | High              | High              | 5        |
| Solar Irrigation                                                | W,E,F           | Medium      | High              | Medium            | 4        |
| Wastewater Reuse (Bahr El-Baqar / Al-Mahsama / New Delta model) | W,E,F           | High        | High              | High              | 5        |
| Wetland Restoration                                             | W,Eco           | Medium      | Medium            | High              | 3        |
| Agro-industrial Efficiency                                      | W,E,F           | High        | High              | Medium            | 4        |

### Ranking

7. Drainage Water Desalination + Solar
8. Wastewater Reuse
9. Agro-industrial Efficiency

## 13. WEFE Project Classification and Bankability Assessment

| Component | Integration Level | Risk Impact   | Bankability Measures                |
|-----------|-------------------|---------------|-------------------------------------|
| Water     | Full              | High positive | Long-term water purchase agreements |
| Energy    | Full              | Medium        | Solar PPA                           |
| Food      | Full              | High          | Farmer contracts                    |
| Ecosystem | Moderate          | Medium        | Carbon credits                      |
| Digital   | Full              | Low           | Smart metering                      |

### Bankability Rating

High

## 14. Integrating WEFE into Existing Projects

Egypt's existing drainage-reuse plants were largely commissioned as conventional treatment facilities and can be retrofitted with WEFE-Nexus features to raise their resilience and revenue profile:

| Action                                       | Timeframe   |
|----------------------------------------------|-------------|
| Solar PV installation at existing/new plants | Short-term  |
| AI optimisation of pumping and treatment     | Short-term  |
| Wetland creation at receiving drains/lakes   | Medium-term |
| Carbon credit mechanism                      | Medium-term |
| Water trading / allocation platform          | Long-term   |

### Expected Benefits

**OPEX Reduction:** 25%

**Energy Cost Reduction:** 40%

**Revenue Increase:** 15%

## 15. Financial and Economic Analysis

### CAPEX

| Component            | Cost (USD million) |
|----------------------|--------------------|
| Desalination Plant   | 350                |
| Solar PV             | 120                |
| Distribution Network | 150                |
| Digital Platform     | 20                 |
| Wetlands             | 30                 |
| Total                | 670                |

### OPEX

Approximately USD 25 million per year.

### Revenue Sources

10. Water tariffs / bulk water purchase agreements
11. Agricultural productivity gains
12. Energy savings from co-located solar generation
13. Carbon credits
14. Green certificates

*CAPEX figures are illustrative and scaled from the order of magnitude of Egypt's existing mega-plants (Bahr El-Baqar and New Delta each represent multi-hundred-million to billion-dollar investment classes); a bankable feasibility study should refine these against site-specific engineering estimates.*

### 15.1 Benchmarking Logic

The USD 670 million envelope used in this report is deliberately positioned as a mid-scale, replicable unit rather than a mega-project on the scale of Bahr El-Baqar or New Delta, both of which required national-level financing structures led by the Armed Forces Engineering Authority. A decentralised project of this size is intended to be financeable by a single multilateral lender or a small blended-finance consortium, and to be repeatable across multiple secondary-drain sites in the Delta without requiring a new national programme each time. When adapting this template to another country or site, the CAPEX table should be built bottom-up from unit costs (USD per m<sup>3</sup>/day of treatment capacity, USD per MW of solar, USD per hectare of distribution network) benchmarked against at least one operating reference project, exactly as this report benchmarks against the three Egyptian plants.

### 15.2 Economic Value Drivers

Beyond the direct revenue lines listed above, the economic case rests on avoided cost: every cubic metre of drainage water reused is a cubic metre that does not need to be drawn from the fixed Nile allocation, and every hectare irrigated from treated drainage water is a hectare of reclaimed desert land added to the cultivated base without competing for Delta 'old land' water rights. A full feasibility study should quantify this

avoided-cost value explicitly, alongside the reduced downstream pollution load into receiving water bodies such as Lake Manzala, since donor climate-finance facilities increasingly score projects on such avoided-externality metrics in addition to direct financial return.

## 16. Capital Stack Design

| Instrument        | Project Stage | Risk Coverage | Expected Benefit        |
|-------------------|---------------|---------------|-------------------------|
| Grant / TA        | Preparation   | High          | Feasibility studies     |
| Concessional Debt | Construction  | Medium        | Lower financing cost    |
| Commercial Debt   | Operation     | Medium        | Scale-up                |
| Equity            | Development   | Medium        | Risk sharing            |
| Blended Finance   | All stages    | High          | Increased bankability   |
| Green Bonds       | Expansion     | Medium        | Institutional investors |

### Proposed Capital Structure

- 15% Grants
- 25% Equity
- 40% Concessional Debt
- 20% Commercial Debt

## 17. Operational Finance Risk Assessment

| Risk Type       | Impact | Mitigation Tool           | Expected Effect |
|-----------------|--------|---------------------------|-----------------|
| Revenue Risk    | High   | Water purchase agreements | Stable cashflow |
| O&M Costs       | Medium | Co-located solar energy   | Reduced costs   |
| Climate Risk    | High   | Insurance                 | Risk transfer   |
| Regulatory Risk | Medium | PPP framework             | Stability       |
| ESG Risk        | Low    | Monitoring systems        | Compliance      |

### Top 5 Risks

15. Water demand fluctuations
16. Energy price volatility
17. Regulatory changes
18. Climate extremes
19. O&M performance

### 17.1 Lessons from the National Reference Programme

Operating experience at Al-Mahsama and Bahr El-Baqar offers two specific, transferable risk lessons. The first concerns hydraulic risk at intake: drainage flow volumes vary seasonally with the agricultural calendar, and pumping-station capacity must be sized with sufficient margin to avoid either starving the plant or requiring costly emergency discharge — a design parameter that should be explicitly tested during feasibility rather than assumed from average annual flow. The second concerns social risk downstream of the intake: shifting drainage water away from informal downstream abstraction points, as documented at Al-Mahsama, can measurably affect smallholders who rely on that water for supplementary irrigation of high-value crops. This is treated in this report as an operational risk with a financial dimension (compensation, grievance-redress costs, reputational risk to lenders) as well as a safeguards issue, and is cross-referenced in Section 20.

## 18. Funding Call Fit Score Assessment

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Example: Green Climate Fund

| Criteria                | Score (/2) |
|-------------------------|------------|
| Nexus Integration       | 2          |
| Expected Outcomes       | 2          |
| Eligibility             | 2          |
| Partnership Feasibility | 2          |
| Budget Alignment        | 2          |
| MRV                     | 2          |
| Readiness               | 2          |

**Total:** 14/14

**Decision:** GO

## 19. Monitoring, Reporting and Verification (MRV)

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### Water

- m<sup>3</sup> treated
- Water quality

### Energy

- MWh produced
- Energy savings

### Food

- Crop yield
- Farmer income

### Ecosystem

- Biodiversity index
- Restored area

## 20. Environmental and Social Safeguards

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### Environmental

- Brine and residual-salt management
- Biodiversity protection in receiving wetlands
- Water quality monitoring

### Social

- Community engagement, including with 'old lands' smallholders affected by drainage reallocation
- Gender inclusion
- Livelihood protection

*Independent research on Al-Mahsama has documented that reallocating drainage water to new-land reclamation can reduce informal downstream access for old-land smallholders who rely on drain water for supplementary irrigation; any new site under this project should include a dedicated smallholder consultation and compensation/access framework from the outset.*

## 21. Expected Impacts

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| Indicator         | Expected Impact                  |
|-------------------|----------------------------------|
| Water Recovered   | 200 million m <sup>3</sup> /year |
| Agricultural Area | 50,000 ha                        |
| Renewable Energy  | 150 MW                           |

| Indicator                 | Expected Impact |
|---------------------------|-----------------|
| CO <sub>2</sub> Reduction | 250,000 t/year  |
| Jobs Created              | 3,000           |
| Beneficiaries             | 500,000 people  |

## 22. Investment Readiness Assessment

| Criterion                | Rating |
|--------------------------|--------|
| Technical Feasibility    | High   |
| Economic Viability       | High   |
| Climate Benefits         | High   |
| Institutional Readiness  | Medium |
| Financial Attractiveness | High   |

### Overall Rating

**Investment Ready, with a Further Feasibility Study Recommended**

## 23. Investor Pitch Summary

**Investment Ask:** USD 670 million

### Why Invest?

- Addresses Egypt's strategic water security challenge under a fixed Nile allocation
- Fully integrated WEFE Nexus project, built on a proven national delivery model (Bahr El-Baqar, Al-Mahsama, New Delta)
- Climate adaptation and mitigation benefits
- Strong revenue generation potential
- Eligible for climate and blended finance
- Replicable across Egypt's Delta drainage network and the wider MENA region

## 24. Conclusions and Recommendations

Egypt is already investing at a scale few countries have attempted in agricultural drainage water reuse: Bahr El-Baqar, Al-Mahsama and New Delta / El-Hammam together represent one of the largest coordinated water-reuse investment programmes in the world, with a combined target contribution on the order of 5 BCM/year by 2026. This project translates that proven national model into a decentralised, solar-integrated, WEFE-Nexus-scored investment opportunity suitable for blended and climate finance, extending the same logic to medium-scale sites across the Delta's secondary drainage network.

## Recommended Next Steps

20. Conduct a full feasibility study for candidate decentralised sites.
21. Develop a PPP structure, informed by the delivery model used for the national mega-plants.
22. Prepare a climate finance concept note.
23. Undertake an Environmental and Social Impact Assessment (ESIA), with explicit attention to old-land smallholder access.
24. Establish an MRV framework aligned with MWRI's Irrigation System 2.0 digital monitoring architecture.
25. Launch an investor engagement process.
26. Submit funding applications to GCF, EIB, EBRD, IsDB and the World Bank.

**Overall WEFE Nexus Score: 95/100 — Excellent Candidate for Climate and Resilience Finance.**

## 25. Using This Template for Other Projects

This closing section is addressed directly to trainees and project teams in the Financing Resilience and WEFE Nexus Professional Series who will use this Egypt report as the model for a new country or site report. The 24 preceding sections form a fixed architecture; what changes from project to project is the evidence populating each section, not the structure itself. The checklist below maps each section to the type of evidence that should replace the Egyptian data.

### 25.1 Section-by-Section Drafting Checklist

| Report Section                                      | What to Replace It With                                                                                                                                                                                                                                     |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–2. Executive Summary & AI Overview                | Restate the country's core water-scarcity driver (fixed allocation, transboundary dependency, aquifer depletion) and score the new project against the same eight criteria (water, energy, food, ecosystem, climate, financial, innovation, replicability). |
| 3–4. Background & Problem Statement                 | Identify at least one operating reference project in-country or in a comparable basin, exactly as this report anchors on Bahr El-Baqar, Al-Mahsama and New Delta — never leave this section purely hypothetical.                                            |
| 5–6. Objectives & National Alignment                | Map objectives to the host country's own national strategy documents (its equivalent of Egypt Vision 2030) so the project reads as policy-aligned, not externally imposed.                                                                                  |
| 7–9. WEFE Analysis, Description, Stakeholders       | Keep the four-sector nexus table format; list real institutional stakeholders by name and mandate, not generic categories.                                                                                                                                  |
| 10–11. Climate Resilience & Logframe                | Retain the 4x4 logframe matrix structure; source climate risk statements from the national climate strategy or NDC.                                                                                                                                         |
| 12–14. Screening, Classification, Retrofitting      | Reuse the scoring matrices unchanged; only the project list and scores should change.                                                                                                                                                                       |
| 15–18. Financials, Capital Stack, Risk, Funding Fit | Build CAPEX bottom-up from a benchmarked reference project; keep the same funding-call scoring rubric (0–2 per criterion, 14 max) for comparability across the training series.                                                                             |
| 19–21. MRV, Safeguards, Impacts                     | Keep indicator categories identical (water, energy, food, ecosystem) so results are comparable across country reports in the series; safeguards must name a real, documented social or environmental risk, not a placeholder.                               |
| 22–24. Readiness, Pitch, Conclusions                | Close with an explicit investment ask and a single composite WEFE Nexus score out of 100, as used throughout this series.                                                                                                                                   |

### 25.2 Standards to Maintain Across the Series

- Every quantitative claim should be traceable to a named, dated source — government statement, multilateral report, or peer-reviewed literature — as in the Sources and References section below.
- Where source figures conflict (as they do for reclamation-area targets across Egyptian sources), state the range and flag it for verification rather than silently picking one number.

- Keep the same visual identity (navy/teal palette, consistent table styling, consistent heading hierarchy) so that all reports in the Financing Resilience and WEFE Nexus Professional Series are immediately recognisable as a single body of work.
- Always close with a Section 25-equivalent note on data provenance and any assumptions a reader should independently verify before using the report for a binding financing decision.

*Prepared by Dr. Hassan Aboelnga as the reference document for the Financing Resilience and WEFE Nexus Professional Series. This structure may be reused for subsequent country and sector reports in the series, subject to the drafting standards set out above.*

## Sources and References

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*This report draws on Egyptian government statements and the following categories of open sources, current as of July 2026: Egyptian Presidency and Cabinet communications on the 8th Cairo Water Week (October 2025) and the Irrigation System 2.0 briefing (August 2025); Ministry of Water Resources and Irrigation materials; CEDARE's Egypt 2030 Shared Water Reuse Strategy; academic literature on the Al-Mahsama drainage reallocation (MDPI, 2023–2024); and engineering-sector reporting on the Bahr El-Baqar and New Delta plants (Arab Contractors, Khatib & Alami, Arab Urban Development Institute). Figures for capacity and reclamation area vary somewhat across sources and should be verified against MWRI's current published data before use in a binding investment document.*